School of Computing

Module Title and Code	AI FOUNDATIONS IN PRACTICE - M33515 - FHEQ 7
Module Coordinator Other lecturers	Dr. Gelayol Golcarenarenji Gelayol.golcarenarenji@port.ac.uk
Assessment Item number	Item2
Assessment Title	Course work (Image data)
Date Issued	2026-03-29

Schedule and Deliverables

Deliverable	Value	Format	Deadline / Date	Late deadline ECF deadline
Course work	50%	A single PDF	2026-05-13 13:00 [GMT/BST]	2026-05-27 13:00 (10 working days after deadline)

Notes and Advice

- The Extenuating Circumstances procedure is there to support you if you have had any circumstances (problems) that have been serious or significant enough to prevent you from attending, completing or submitting an assessment on time. If you complete an Extenuating Circumstances Form (ECF) for this assessment, it is important that you use the correct module code, item number and deadline (not the late deadline) given above.
- ASDAC are available to any students who disclose a disability or require additional support for their academic studies with a good set of resources on the ASDAC moodle site
- The University takes any form of academic misconduct (such as plagiarism or cheating) seriously, so please make sure your work is your own. Please ensure you adhere to our *Student Conduct Policy* and watch the video on Plagiarism.
- Any material included in your coursework should be fully cited and referenced in **APA 7** format. Detailed advice on referencing is available from the library, also see TECFAC 08 Plagiarism.
- Any material submitted that does not meet format or submission guidelines, or falls outside of the submission deadline could be subject to a cap on your overall result or disqualification entirely.
- If you need additional assistance, you can ask your personal tutor, student engagement officer ana.baker@port.ac.uk, academic tutor eleni.noussi@port.ac.uk or your lecturers.

- If you are concerned about your mental well-being, please contact our Well-being service

AI Statement

AI tools, including Generative AI, **should not be used to**:

• generate content to include in your assessment submission

• generate code to include in your assessment submission

AI tools, including Generative AI, **can be used to**:

• develop your understanding of the concepts covered in the module

• practice use of analytics and programming skills

• check/improve the clarity of your language

Use AI tools critically. They make mistakes and their language is not precise. For example, accuracy and precision may be used as synonyms, while in machine learning these are metrics with very specific meaning and using them interchangeably is not appropriate.

Coursework 2

Please only use your ID anywhere on the coursework or the cover sheet.

Accurate recognition of traffic lights is critical for Autonomous Vehicles (AVs) to avoid crashes due to red light running which leads to reducing deaths and injuries of people. Hence, in this project, we are looking for an image classification system to classify two classes of traffic lights red and green with high accuracy. Please write a report in 1500 words explaining different stages of achieving the goal of this project. The report should include the following steps in detail. The dataset has been provided for you on Moodle (50%).

2.1. Prepare the dataset for training (Data augmentation, splitting the dataset in training and

testing datasets…).

2.2. Design a 2D convolutional neural network to classify these images.

2.3. Provide the training loss and accuracy plots and report the accuracy of the model.

2.4. Select five testing images (The images should not be in your training dataset). Please report the predicted labels for all these selected images.

2.5. Please provide two ways (with evidence) to further improve the accuracy of your classifier.

Grading Criteria

The grading criteria (in 100% breakup of marks) is as follows:

60% - Thorough analysis and evaluation of results.

20% - Use of tables and figures when reporting the results.

10% - Concise and clear organization and presentation of findings, clear language and

writing style.

10% -Provide ways of further improving the accuracy of your developed models

Deliverables

Please submit the followings on Moodle in the designated folder (Coursework submission folder):

1. A final report in PDF format with 1500 words excluding figures and tables.
2. The source codes (. ipynb files).